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ENCI335 Structural Analysis & Systems I (II) Analysis of Indeterminate Structures (2) Methods for Statically Indeterminate Frames

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What are (Structural) Frames?

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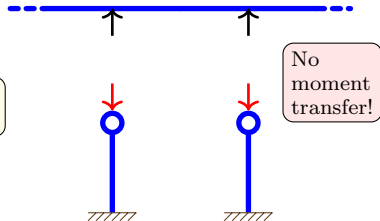
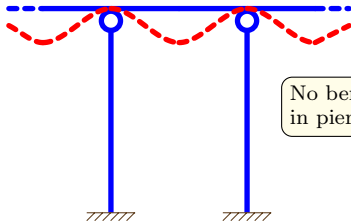
Area

Examples

Structural frames are structural systems made of beams and columns connected with **moment**, shear, and axial force transfer.



← This is **NOT** a frame!



What are (Structural) Frames?

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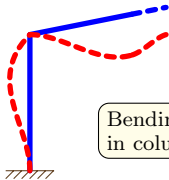
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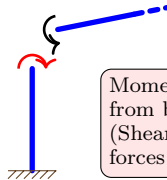
Structural frames are structural systems made of beams and columns connected with **moment**, shear, and axial force transfer.



This is a frame!



Bending
in columns!



Moment is transferred
from beam to column!
(Shear and axial
forces not drawn)

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Stability of Portal Frames

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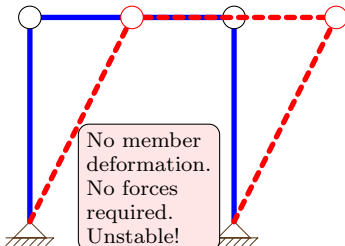
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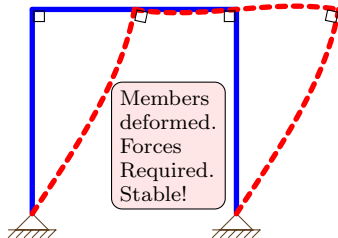
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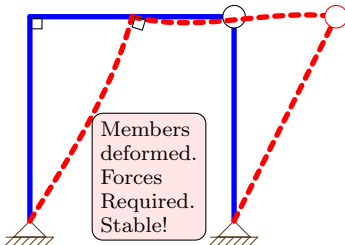


$$SI = 4 - 5 = -1$$



$$SI = 4 - 3 = 1$$

Statically Indeterminate!



$$SI = 4 - 4 = 0, \text{ statically determinate!}$$

For Portal Frames:

$$SI = n_R - n_E$$

Axial deformations
in frame elements
are often negligible.

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Statically Determinate Frames

Statically Determinate Frames - Example 1 - 1/4

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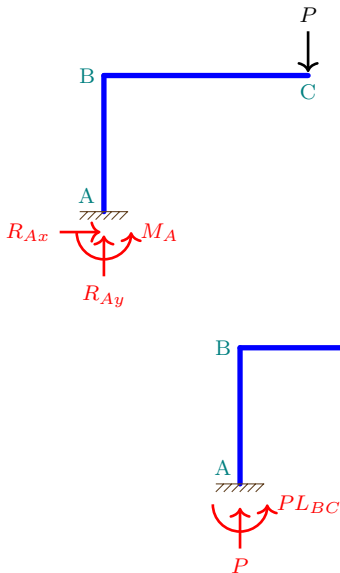
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Step 1: Label and determine support reactions using the global equilibrium equations:

$$\sum F_x = 0 \implies R_{Ax} = 0$$

$$\sum F_y = 0 \implies R_{Ay} = P$$

$$\sum M@A = 0 \implies M_A = PL_{BC}$$

For a statically determinate frame, all reactions can be readily calculated.

Statically Determinate Frames - Example 1 - 2/4

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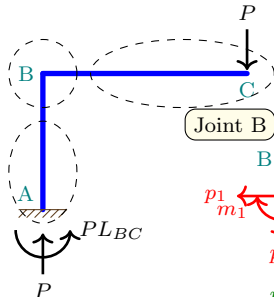
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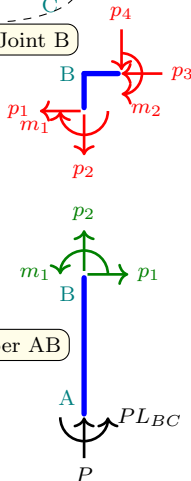
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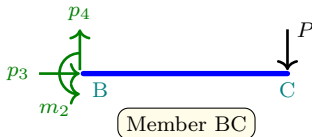


Joint B



Member AB

Step 2: Draw free-body diagrams. One for each member and one for each joint.



Member BC

Joint B, which is a point, with only two members connected and no external point loads and point moments applied to can often be skipped because the transfer of moments and forces between beams and columns is quite straightforward.

Statically Determinate Frames - Example 1 - 3/4

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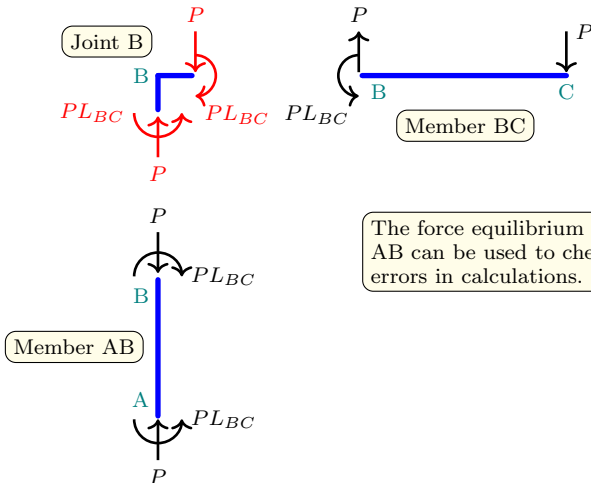
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Step 3: Use the equilibrium equations to determine $p_1, p_2, p_3, p_4, m_1, m_2$.



The force equilibrium at member AB can be used to check for any errors in calculations.

Statically Determinate Frames - Example 1 - 4/4

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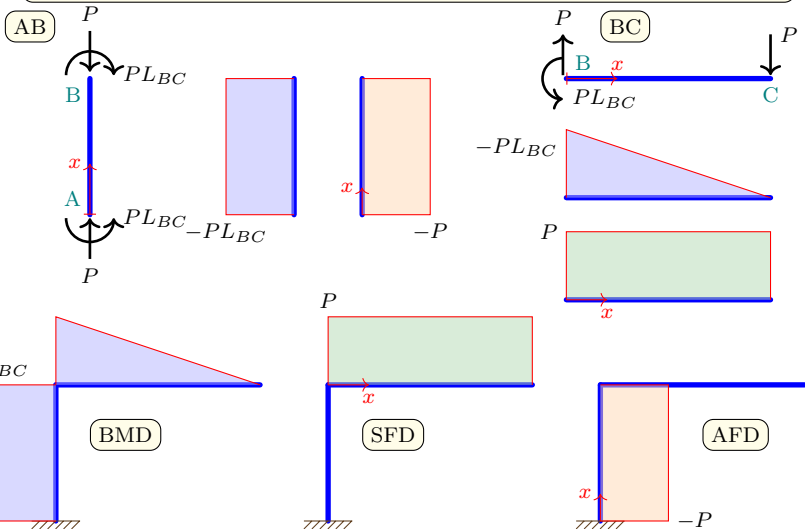
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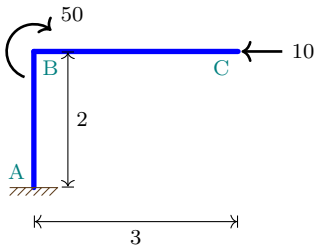
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Step 4: Draw the BMD, SFD, and AFD for the entire structure by combining the individual member diagrams.



- When a free-body diagram of a frame member is created from a section cut, the unknown internal forces are exposed.
- For a frame member resisting both flexural and axial deformation, these internal forces include the bending moment, shear force, and axial force.
- While M , V , and N are standard labels, it is often more general to use p and m to treat them as generic point loads and moments during intermediate analysis steps. This avoids immediate concern regarding their specific orientation as bending, shear, or axial components.
- When possible, determine unknown internal forces by applying equilibrium equations to each member's free-body diagram, incorporating known applied loads and support reactions.
- Transfer these internal forces to the joints with the same magnitude but in the opposite direction (action-reaction).
- Solve for any remaining unknowns or check them using the equilibrium equations at the joints.
- At a beam-column joint connecting only two members with no external loads, internal forces are transferred directly without a change in magnitude. Consequently, the free-body diagram of such a joint is often omitted.
- Once all the member internal forces are obtained, the BMD, SFD, and AFD of each member can be readily drawn, and combined for the entire structure.
- SFD and AFD depend on the orientation of each individual member. Therefore, the member x -axis should be clearly indicated on each member.

Statically Determinate Frames - Example 2



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Statically Determinate Frames - Example 3

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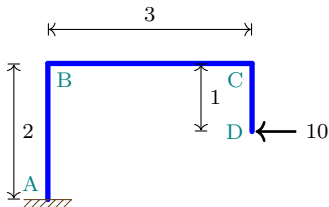
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Statically Determinate Frames - Example 4 (Portal Frame)

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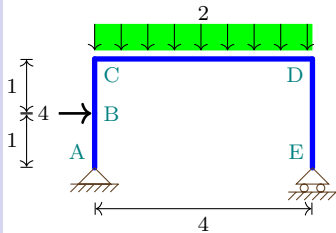
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Statically Determinate Frames - Example 5 (Internal Hinges)

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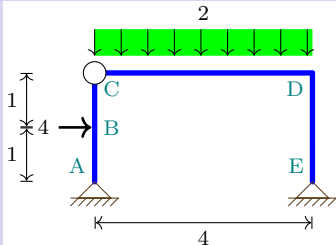
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Statically Determinate Frames - Example 6 (> 2 members at a joint)

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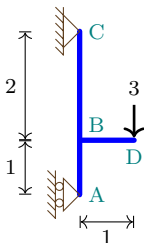
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Statically Determinate Frames - Example 7 (3D Beam)

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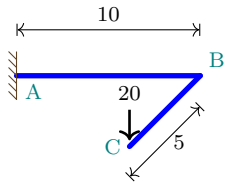
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Degree of Statical Indeterminacy

Degree of Static Indeterminacy for Frames

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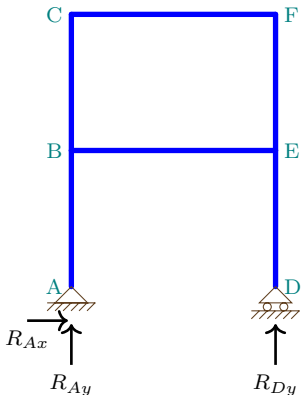
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- The equation $SI = n_R - n_E$ is only applicable to beams, but **it is not generally true for frames**, particularly when there are joints with more than 2 members connecting to them.



The 3 reactions can be obtained from the global equilibrium equations. The BMD, SFD, and AFD of members AB and DE can be drawn.

However, the member internal forces of all other members cannot be determined without using additional information.

For example, it is not clear how the forces of member AB acting on joint B are distributed to members BC and BE.

We need a better formula for SI of a frame.

We will come back again to check the SI of this frame.

Degree of Static Indeterminacy SI for 2D Frames

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- How many unknowns in a frame?
 - Total number of reactions n_R
 - Each frame member has 3 unknown member forces: M , V , and N .
 - Each truss member has 1 unknown: N .
- How many equilibrium equations are available?
 - Each joint with at least one frame member connected to it has 3 joint equilibrium equations: $\sum F_X = 0$, $\sum F_Y = 0$, and $\sum M_Z = 0$.
 - Each joint with only truss members connected to it has 2 joint equilibrium equations: $\sum F_X = 0$ and $\sum F_Y = 0$.
 - Each internal hinge provides one additional equilibrium equation.

- The equation for SI is

$$SI = n_R + n_{MF} - n_{JE} - n_H \quad (1)$$

where n_{MF} , n_{JE} , and n_H are the total numbers of member forces, equilibrium equations, and internal hinges, respectively. **Note that n_H is now expressed explicitly without being lumped into n_{JE} .**

- The equation can be rewritten as

$$SI = n_R + 3(n_M - n_J) - n_H \text{ (Frames without truss members)} \quad (2)$$

$$SI = n_R + n_M - 2n_J \text{ (Truss structures)} \quad (3)$$

where n_M and n_J are the total numbers of members and joints, respectively.

- The equation $SI = n_R + n_{MF} - n_{JE} - n_H$ is the most general formula for determining SI . The other two equations are derived by simplifying this expression to suit specific structures.
- When there is a mixed use of member types (frame and truss members) in a structure, it is better to use the general equation to determine SI .
- The number of member forces (n_{MF}) is 1 for each truss member, and 3 (2D) or 6 (3D) for each frame member.
- The number of equilibrium equations (n_{JE}) is 3 (2D) or 6 (3D) for each joint with at least one frame member connected to it, and 2 (2D) or 3 (3D) for each joint with only truss members connected to it.

SI Example 1

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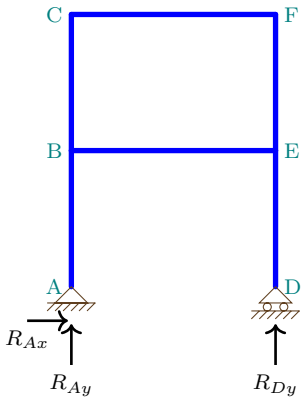
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SI Examples 2 and 3

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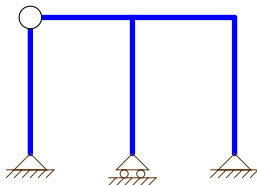
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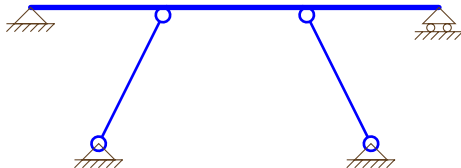
Moment-
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Examples



$$\begin{aligned} n_R &= 5 \\ n_M &= 5 \\ n_J &= 6 \\ n_H &= 1 \end{aligned}$$

$$\begin{aligned} SI &= n_R + 3(n_M - n_J) - n_H \\ &= 5 + 3(5 - 6) - 1 = 1 \end{aligned}$$



$$\begin{aligned} n_R &= 7 \\ n_{MF} &= 3 \times 3 + 2 \times 1 = 11 \\ n_{JE} &= 4 \times 3 + 2 \times 2 = 16 \end{aligned}$$

$$\begin{aligned} SI &= n_R + n_{MF} - n_{JE} \\ &= 7 + 11 - 16 = 2 \end{aligned}$$

Degree of Static Indeterminacy SI for 3D Frames

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Examples

- How many unknowns in a frame?
 - Total number of reactions n_R
 - Each frame member has 6 unknown member forces: T_x , M_y , M_z , V_y , V_z and N .
 - Each truss member has 1 unknown: N .
- How many equilibrium equations are available?
 - Each joint with at least one frame member connected to has 6 joint equilibrium equations: $\sum F_X = 0$, $\sum F_Y = 0$, $\sum F_Z = 0$, $\sum M_X = 0$, $\sum M_Y = 0$, and $\sum M_Z = 0$.
 - Each joint with only truss members connected to has 3 joint equilibrium equations: $\sum F_X = 0$, $\sum F_Y = 0$ and $\sum F_Z = 0$.
 - Each internal hinge (one for each axis) provides one more equilibrium equation.
- The equation of SI is still the same

$$SI = n_R + n_{MF} - n_{JE} - n_H \quad (4)$$

- The equation can be rewritten as

$$SI = n_R + 6(n_M - n_J) - n_H, \text{ (Frames without truss members)} \quad (5)$$

$$SI = n_R + n_M - 3n_J, \text{ (Truss structures)} \quad (6)$$

SI Examples 4 and 5 - 3D Frames

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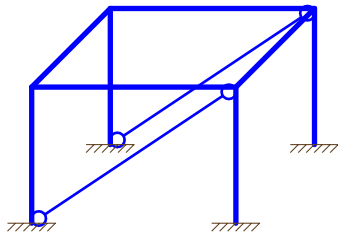
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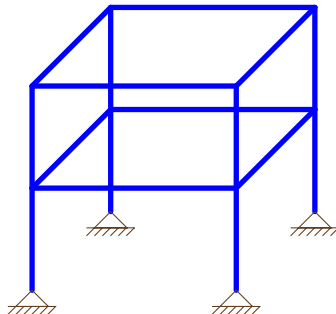
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Examples



$$\begin{aligned}n_R &= 4 \times 6 = 24 \\n_{MF} &= 8 \times 6 + 2 = 50 \\n_{JE} &= 8 \times 6 = 48\end{aligned}$$

$$\begin{aligned}SI &= n_R + n_{MF} - n_{JE} \\&= 24 + 50 - 48 = 26\end{aligned}$$



$$\begin{aligned}n_R &= 4 \times 3 = 12 \\n_M &= 16 \\n_J &= 12\end{aligned}$$

$$\begin{aligned}SI &= n_R + 6(n_M - n_J) \\&= 12 + 6 \times (16 - 12) = 36\end{aligned}$$

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Qualitative Sketch

Deflected Shapes of Frames

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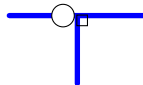
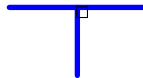
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- Like beams, frames deflect in smooth curves, but they can also move laterally (i.e. **sway**).
- At supports and connections:



Deflected Shapes and BMD of Frames

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For deflection:

- Frame members are assumed axially **rigid**, because axial deformation is typically orders of magnitude smaller than flexural deformation and thus negligible.
- Start from the loading points.
- Rotate the joints as necessary to facilitate deflection (check the angles).
- Identify the tension side of the members.
- Locate the points of contraflexure (PCs).
- Complete the deflected curves.

For BMD:

- At PCs, BM is zero and changes from hogging to sagging, and vice versa.
- Draw the BM on the tension side.
- Ensure that the BMs at the supports are consistent with the support types.
- At joints with only 2 members connected, ensure that the BMs of the two members are equal to each other at the joint.
- At joints with > 2 members connected, ensure that the sum of BMs at the joint is zero.

Deflected Shapes and BMD of Frames - Examples

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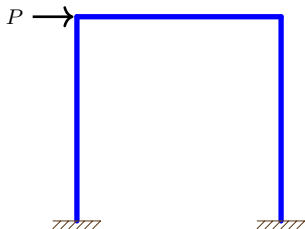
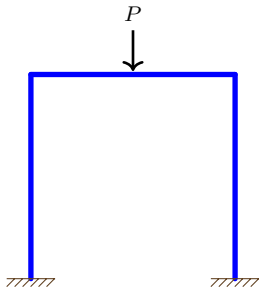
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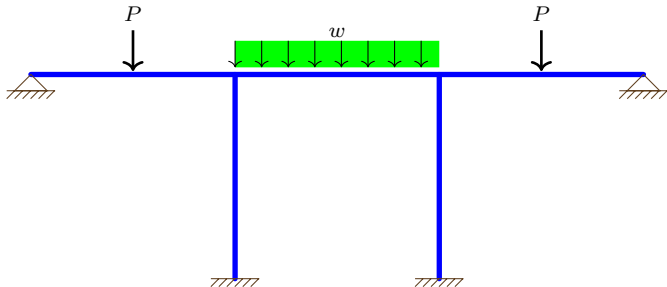
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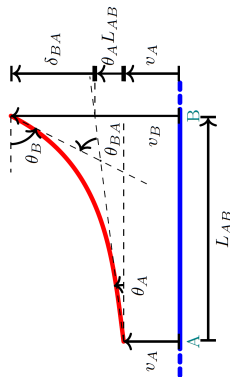
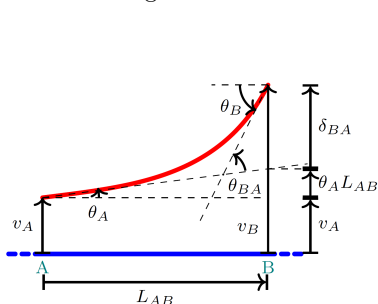
**Moment-
Area**

Examples

Moment-Area Theorems

Moment-Area Theorems for Columns - 1/3

- The principles of the moment-area theorems discussed for beam analysis also apply to frame analysis.
- The column response in frames can be viewed as a beam response rotated 90 degrees.



- However, this interpretation requires modification.

Moment-Area Theorems for Columns - 2/3

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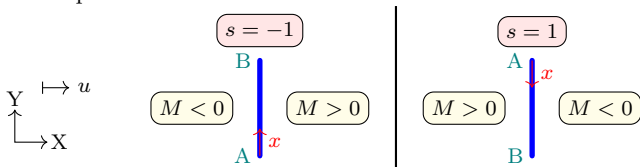
Moment-Area

Examples

- The terms $\theta_A L_{AB}$ and δ_{BA} represent local displacements relative to the orientation of the frame member.
- Therefore, when relating these to the lateral displacements of columns (i.e., u), care must be taken to ensure the transformation from local to global coordinate systems is consistent with sign conventions.
- Transforming displacements between local and global coordinate systems often involves cos or sin.
- Since column members are always vertical, the transformation can be simplified by introducing an additional sign parameter, s , instead of using trigonometric functions.
- The equation for the lateral displacement of columns becomes

$$u_B = u_A + s(\theta_A L_{AB} + \delta_{BA}) \quad (7)$$

where s depends on the orientation of the local x -axis of the column:



Moment-Area Theorems for Columns - 3/3

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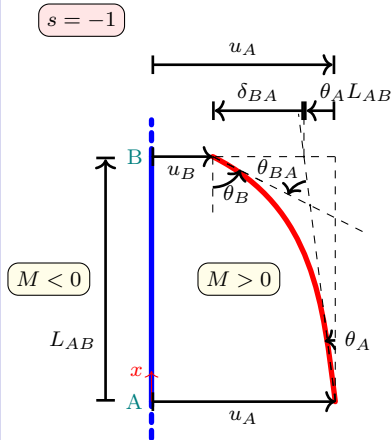
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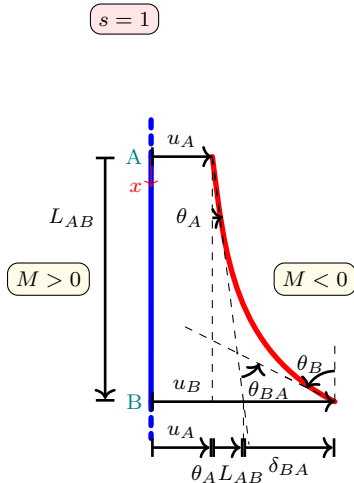
Moment-Area

Examples



$$\theta_B = \theta_A + \theta_{BA}$$

$$u_B = u_A + (-1) \cdot (\theta_A L_{AB} + \delta_{BA})$$



$$\theta_B = \theta_A + \theta_{BA}$$

$$u_B = u_A + 1 \cdot (\theta_A L_{AB} + \delta_{BA})$$

- When drawing the bending moment diagram (BMD) for columns, the shape of the diagram remains unchanged regardless of the orientation of the column's x -axis.
- However, the sign of the moment M depends on the chosen orientation.
- In the calculation of tangential deviation δ_{BA} , it is critical to distinguish between the positive and negative regions of the BMD area.
- Therefore, it is helpful to indicate the orientation of the local x -axis when computing the tangential deviation.

Moment-Area Theorems for Frame Analysis - Example 1

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Stability

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3D Frame

Sketch

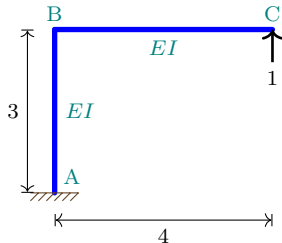
Examples

Moment-

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Examples

Determine the displacements (u , v , θ) at the joints B & C, and draw the deflected shape.



Moment-Area Theorems for Frame Analysis - Example 2

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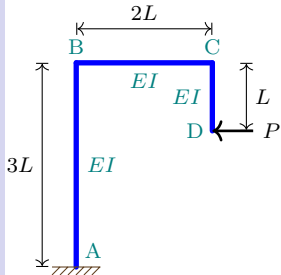
Examples

Moment-

Area

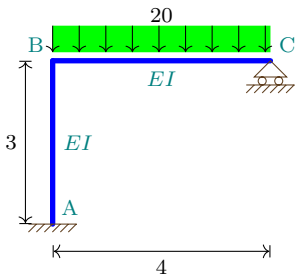
Examples

Determine the displacements (u , v , θ) at the joints B, C & D, and draw the deflected shape.



Moment-Area Theorems for Frame Analysis - Example 3

Determine the displacements (u , v , θ) at the joints B & C, and draw the final BMD and deflected shape.



Moment-Area Theorems for Frame Analysis - Example 4

Determine the displacements (u , v , θ) at the joints B, C & D, and draw the deflected shape, and the final BMD, SFD, and AFD.

